(i)

When n=1:

$$p_1(x) = (x+1)^2 - 3x = x^2 - x + 1$$
$$\implies (x+1)p_1(x) = (x+1)(x^2 - x + 1) = x^3 + 1 = (x+1)q_1(x)$$
$$\implies p_1(x) = q_1(x)$$

When n=2:

$$p_2(x) = (x+1)^4 - 5x(x^2 + x + 1)$$
$$\implies p_2(x) = (x^4 + 4x^3 + 6x^2 + 4x + 1) - (5x^3 + 5x^2 + 5x) = x^4 - x^3 + x^2 - x + 1$$
$$\implies (x+1)p_2(x) = (x+1)(x^4 - x^3 + x^2 - x + 1) = x^5 + 1 = (x+1)q_2(x)$$
$$\implies p_2(x) = q_2(x)$$

When n=3:

$$p_3(x) = (x+1)^6 - 7x(x^2 + x + 1)^2$$
$$\implies p_3(x) = (x^6 + 6x^5 + 15x^4 + 20x^3 + 15x^2 + 6x + 1) - (7x^5 + 14x^4 + 21x^3 + 14x^2 + 7x)$$
$$\implies p_3(x) = x^6 - x^5 + x^4 - x^3 + x^2 - x + 1$$
$$\implies (x+1)p_3(x) = (x+1)(x^6 - x^5 + x^4 - x^3 + x^2 - x + 1) = x^7 + 1 = (x+1)q_3(x)$$
$$\implies p_3(x) = q_3(x)$$

When n=4:

$$p_4(x) = (x+1)^8 - 9x(x^2 + x + 1)^3$$

Let x=1:

$$p_4(1) = 2^8 - 9 \cdot 1 \cdot 3^3 = 256 - 243 = 13$$
$$q_4(1) = 1^8 - 1^7 + 1^6 - 1^5 + 1^4 - 1^3 + 1^2 - 1 + 1 = 1$$
$$\implies p_4(1) \neq q_4(1)$$
$$\implies p_4(x) \neq q_4(x)$$

(ii)(a)

$$\begin{aligned} q_1(300) = \frac{300^3 + 1}{301} &= (301)^2 - 3(300) \\ &= 301^2 - 30^2 \\ &= (301 - 30)(301 + 30) \\ &= 331 \times 271 \end{aligned}$$

(ii)(b)

$$\begin{aligned}\frac{7^{49}+1}{7^7+1} = \frac{(7^7)^7+1}{7^7+1} &= q_3(7^7) = p_3(7^7) \\ &= (7^7+1)^6 - 7\cdot 7^7(7^{14}+7^7+1)^2 \\ &= \Big((7^7+1)^3\Big)^2 - \Big(7^4(7^{14}+7^7+1)\Big)^2 \\ &= \Big((7^7+7^0)^3 + 7^4(7^{14}+7^7+7^0)\Big)\Big((7^7+1)^3 - 7^4(7^{14}+7^7+7^0)\Big)\end{aligned}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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